

Pregnancy Success Classification Using Infertility Treatment Data

LG AI Hackathon

* CONTENTS

1. Problem Statement & Dataset Overview

2. Overall Workflow

3. Data Preprocessing

- EDA & PREPROCESSING
- FEATURE ENGINEERING
- ENCODING

4. Model Training & Optimization

- K-FOLD CROSS VALIDATION
- MODEL 고려 및 SEED ENSEMBLE
- HYPERPARAMETER TUNING

5. Final Model Selection

* PROBLEM STATEMENT & DATASET OVERVIEW

Topic & Background

To reduce the treatment burden for patients undergoing infertility procedures while enabling medical institutions to offer differentiated services, we **developed an AI model that predicts pregnancy success** and identifies the optimal features influencing pregnancy outcomes using real-world infertility patient data.

Dataset Overview

- Train Data: Infertility treatment records of 256,351 samples, each with a unique ID, containing pregnancy success labels and 68 variables.
- Test Data: Infertility treatment records of 90,067 samples, each with a unique ID, containing 67 variables (labels excluded).

Age at treatment, Treatment type,
Total number of pregnancies, Total number of treatments,
Number of embryos transferred, Source of Sperm
시술 당시 나이, 시술 유형, 총 임신 횟수, 총 시술 횟수, 이식된 배아 수, 정자 출처 등..

* OVERALL WORKFLOW

Main Function Workflow

```
def main(config_params, model_param_dict):
    start_time = time.time() # 시작 시간 기록
    y_probs = []
    for seed in config_params['seed_list']:
        print('\n'+'*'*30)
        print(f"Seed Number : {seed}")
        print(""*30)
        for model_name, model_params in model_param_dict.items():
            print(f"<Model : {model_name}>")
            # seed 고정
            set_seed(seed)

            # 데이터 셋 읽기
            df_train, df_test, df_submit = read_data(config_params['data_path'])

            # 데이터 전처리
            df_train = feature_engineering(df_train)
            df_test = feature_engineering(df_test)
            base_features, cat_features, num_features, removal_features = make_feature_lists(df_train)

            # 범주형 변수 인코딩
            df_train, df_test = encoding_cat_features(df_train, df_test,
            | | | | | | | | | | cat_features, config_params, model_name, seed)

            # 결측치 처리
            df_train = filling_missing_values(df_train, cat_features, num_features, model_name)
            df_test = filling_missing_values(df_test, cat_features, num_features, model_name)

            # train 데이터 학습
            models = model_kfold(df_train, base_features, cat_features, config_params,
            | | | | | | | | | | model_params, model_name, seed)

            # 제출 파일 생성
            y_prob = predict_test(models, df_test, base_features)
            y_probs.append(y_prob)

    make_submission(y_probs, df_submit)
```

시드 고정 (set_seed)

데이터 셋 읽기 (read_data)

데이터 전처리 및 파생변수 생성 (feature_engineering)

범주형 변수 인코딩 (encoding_cat_features)

결측값 처리 (filling_missing_values)

train 데이터 학습 및 k-fold 교차검증 (model_kfold)

Generate Predictions 및 제출파일 출력 (predict_test)

Selected XGB model →

5 seed ensemble + hyperparameter tuning

* OVERALL WORKFLOW

_ Main Function Definition

set_seed

Fix seeds for multiple libraries

read_data

Load train, test, and sample_submission data files

feature_engineering

Perform feature engineering based on EDA results

make_feature_lists

Create feature lists (numeric, categorical, excluded) as needed

encoding_cat_features

Select encoder by model type

filling_missing_values

Handle missing values

model_kfold

Train model with K-Fold Cross-Validation

predict_test

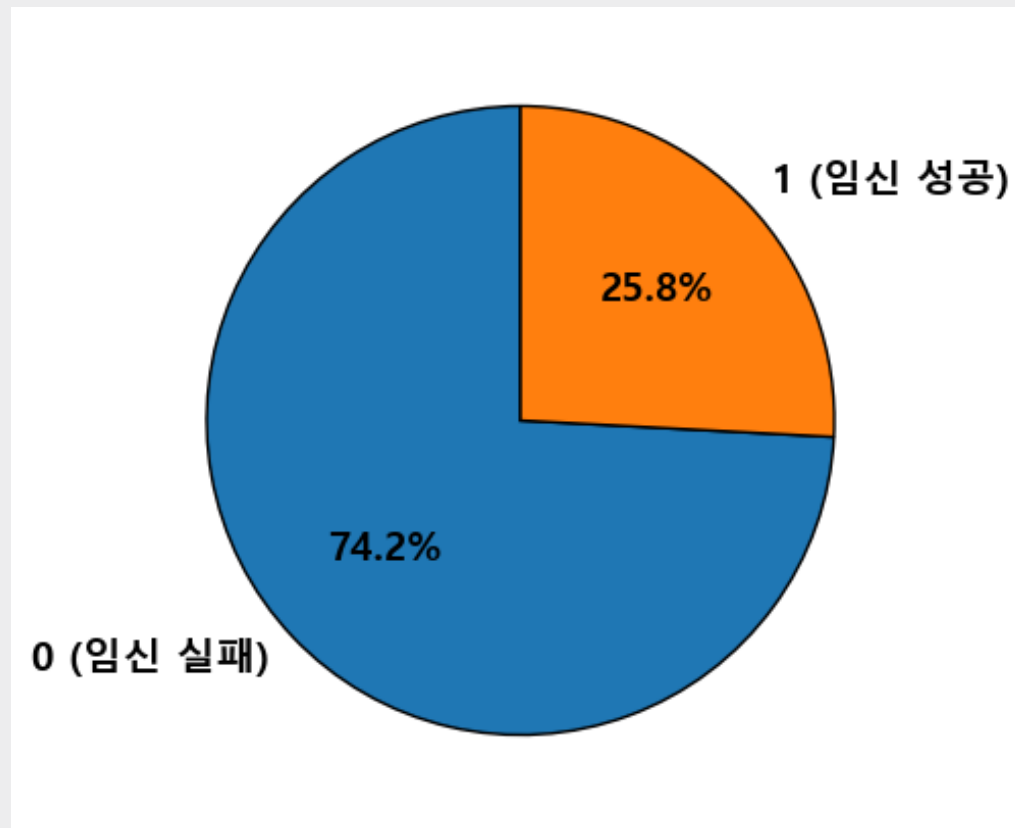
Infer test data with each fold model and use the average as the prediction value

make_submission

Calculate final probability as the average of “pregnancy success probabilities” across seeds and generate the submission file

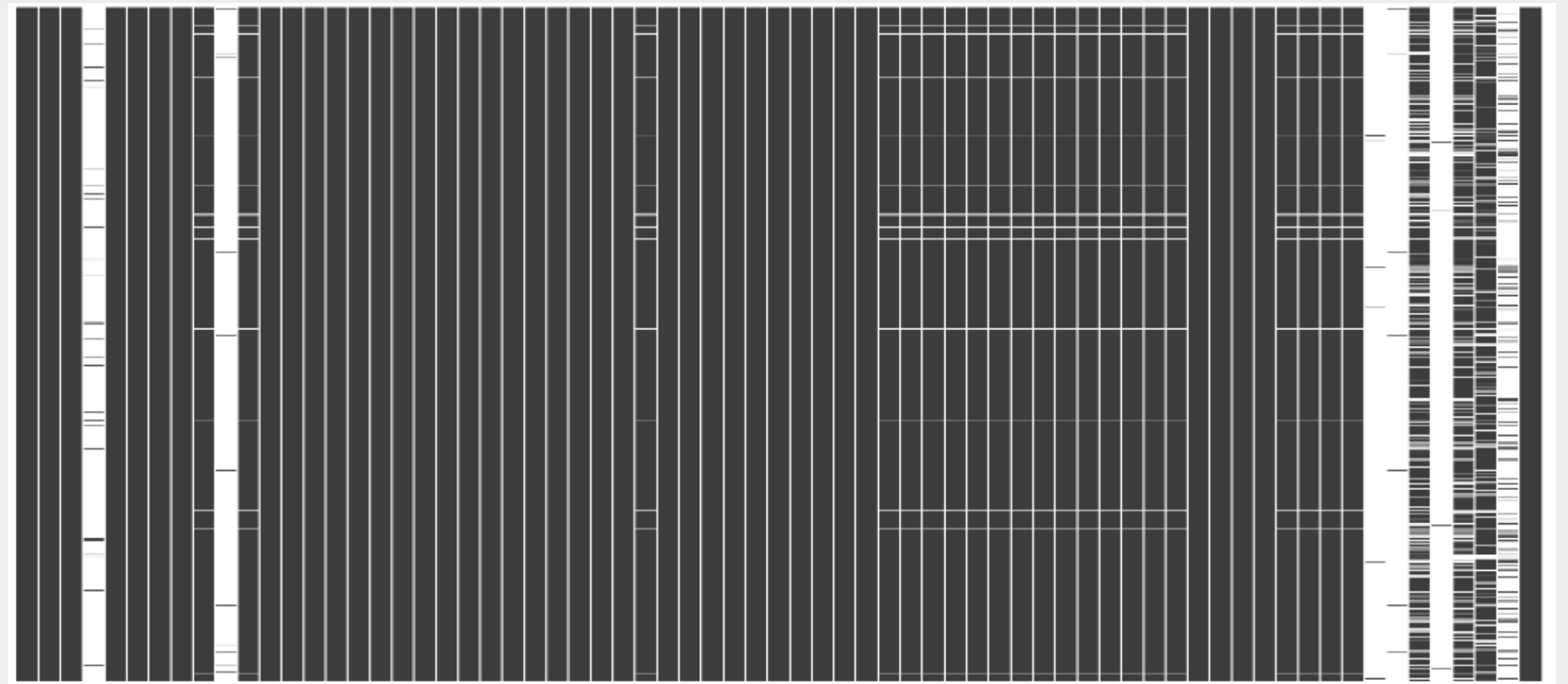
* EDA & PREPROCESSING

Data Imabalance



target variable (0/1) is imbalanced.

Check Missing Value Distribution



Distribution of missing values varies by treatment type.
ex. Variables related to IVF procedures (oocyte retrieval and mixing, embryo transfer) are missing for DI treatments, as these steps are not performed.
(난자 채취 및 혼합, 배아 이식 등 IVF 관련 변수의 경우, DI 시술에서는 진행되지 않는 과정이므로 결측)

* EDA & PREPROCESSING

Missing value handling

IVF-related variables in DI data DI 데이터의 IVF 관련 변수

For DI procedures, missing values were considered as not collected and replaced with -1.

착상 전 유전 검사 사용 여부, PGS 시술 여부, PGD 시술 여부

'Preimplantation Genetic Testing Use', 'PGS Procedure Use', 'PGD Procedure Use'

For columns where all non-missing rows have a value of 1, missing values were replaced with 0 (these columns are IVF-related, so DI data was replaced with -1).

Elapsed-day related columns 경과일 관련 칼럼

Missing values caused by unperformed procedures were considered and replaced with -1.

임신 시도 또는 마지막 임신 경과 연수, 특정 시술 유형

'Pregnancy Attempt or Time Since Last Pregnancy', 'Specific Treatment Type'

Other uninterpretable columns were replaced with -1 or UNK based on experiments showing the best performance.

* EDA & PREPROCESSING

Missing value handling

In conclusion, missing values in the IVF data columns 'Preimplantation Genetic Testing Use', 'PGS Procedure Use', 'PGD Procedure Use' were replaced with 0, while all other missing values were replaced with -1.

```
# 확정 결측치 처리
for feat in ['PGS 시술 여부', 'PGD 시술 여부', '착상 전 유전 검사 사용 여부']:
    ivf_df[feat] = ivf_df[feat].fillna(0)
```

```
✓ def filling_missing_values(df_input, cat_features, num_features, model):
    df = df_input.copy()

    ✓ if model == 'CB':
    ✓     for cat_feature in cat_features:
        df[cat_feature] = df[cat_feature].fillna('UNK')
        df[cat_feature] = df[cat_feature].astype(str)

    ✓ for num_feature in num_features:
        df[num_feature] = df[num_feature].fillna(-1)

    return df
```

* EDA & PREPROCESSING

Variable Type Conversion

Age and count-related variables were converted to numeric types.

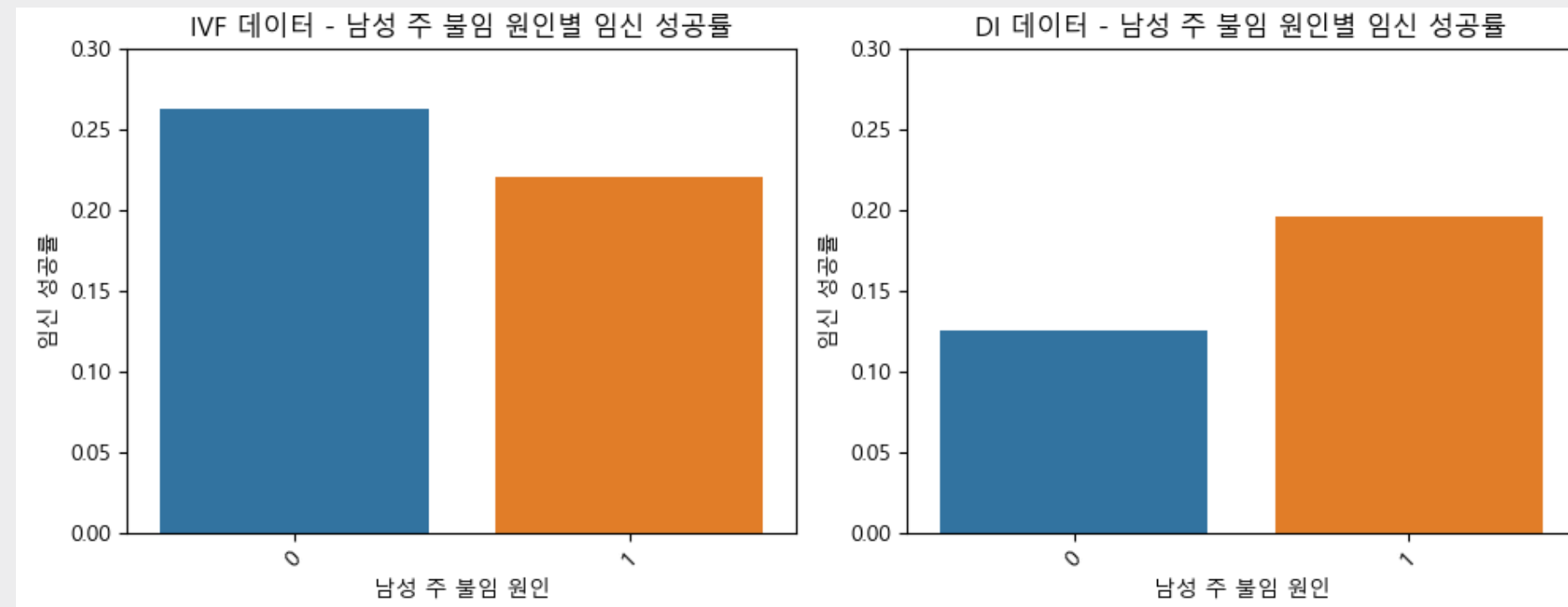
```
# '시술 당시 나이' 변수 수치형으로 변환
df['시술 당시 나이'] = df['시술 당시 나이'].map({
    '만18-34세':1, '만35-37세':2, '만38-39세':3, '만40-42세':4,
    '만43-44세':5, '만45-50세':6, '알 수 없음':7
})

# '~횟수' 변수 수치형으로 변환
count_features = [feat for feat in df.columns if '횟수' in feat]
for feat in count_features:
    df[feat] = df[feat].apply(lambda val : int(val[0])) # n회/n회 이상 (str) -> n (int)

# '정자, 난자 기증자 나이' 변수 수치형으로 변환
for feat in ['난자 기증자 나이', '정자 기증자 나이']:
    df[feat] = df[feat].apply(lambda val : int(val[1:3]) if val[0] == '만' else -1)
```

* EDA & PREPROCESSING

DI Data Flip



Example visualization: pregnancy success rate by male primary infertility cause.

EDA revealed that the target distributions differ between IVF and DI treatments for some infertility cause variables.

→ For these infertility cause columns showing this pattern, **DI data values were flipped (0 ↔ 1)**.

→ This adjustment helps balance the minority class DI treatment data in the symmetric XGBoost model.

- flip 변수 : 남성 주 불임 원인, 남성 부 불임 원인, 여성 주 불임 원인, 여성 부 불임 원인, 부부 주 불임 원인, 부부 부 불임 원인, 불임 원인 - 정자 농도
- Flip variables include: 'Primary Male Infertility Cause', 'Secondary Male Infertility Cause', 'Primary Female Infertility Cause', 'Secondary Female Infertility Cause', 'Primary Couple Infertility Cause', 'Secondary Couple Infertility Cause', 'Infertility Cause - Sperm Concentration'

* FEATURE ENGINEERING

Variable Removal

'Fresh Embryo Use', 'Frozen Embryo Use'

A discrepancy was observed between the distributions of the related variables 'Total Embryos Created' and 'Frozen-Thawed Embryos'.
⇒ Retained 'Total Embryos Created' and 'Frozen-Thawed Embryos', which have more information, and removed other variables.

총 생성 배아 수 > 0	신선 배아 사용 여부	개수	해동된 배아 수 > 0	동결 배아 사용 여부	개수
0	0	46161	0	0	215966
0	1	13479	1	0	259
1	0	54	1	1	40126
1	1	196657			

Columns with identical values

If all rows have the same value, the variable is removed.

```
removal_features = {'신선 배아 사용 여부', '동결 배아 사용 여부'}  
removal_features |= set(df.columns[df.nunique(dropna=False) == 1])
```

* FEATURE ENGINEERING

Derived Variable Creation

‘Primary Reason for Embryo Creation’

To clearly represent cases with multiple selections, dummy columns were created.

```
# 배아 생성 주요 이유 다중 선택 변환 (4개 더미 컬럼 생성)
categories = ['현재 시술용', '배아 저장용', '기증용', '난자 저장용']
for category in categories:
    df[f'배아 이유_{category}'] = df['배아 생성 주요 이유'].apply(lambda x: 1 if pd.notna(x) and category in x else 0)
```

‘Live Birth Success Rate’ ‘Total Number of Births’ / ‘Total Number of Pregnancies’

‘Pregnancy Success Rate’ ‘Total Number of Pregnancies’ / ‘Total Number of Treatments’

```
# '출산 성공률', '임신 성공률' 파생변수 생성
df['출산 성공률'] = df['총 출산 횟수'] / df['총 임신 횟수']
df['임신 성공률'] = df['총 임신 횟수'] / df['총 시술 횟수']
```

* ENCODING_CATEGORY FEATURE

Encoder settings

After comparing encoder selection experiments for each model—XGB, LGBM, and CatBoost—the following encoders were adopted per model:

- label encoding
- ordinal encoding
- target encoding
- catboost encoding
- frequency encoding

The encoders specified in the config_params were used according to the model.

- CatBoost – its own built-in Encoder
- XGB – ordinal Encoder
- LGBM – targetEncoder

* K-FOLD CROSS VALIDATION

K-FOLD Cross-Validation

Model training and experiments were conducted using K-fold cross-validation.

Considering **target imbalance**, data was split using **Stratified K-fold** to maintain the same target distribution in each fold.

K-fold comparison experiments

5, 7, 9, 11 k values were varied and compared, resulting in selecting 5 folds $\Rightarrow$ 'k-fold': 5

After splitting the data into 5 folds, each fold was used once as the validation set

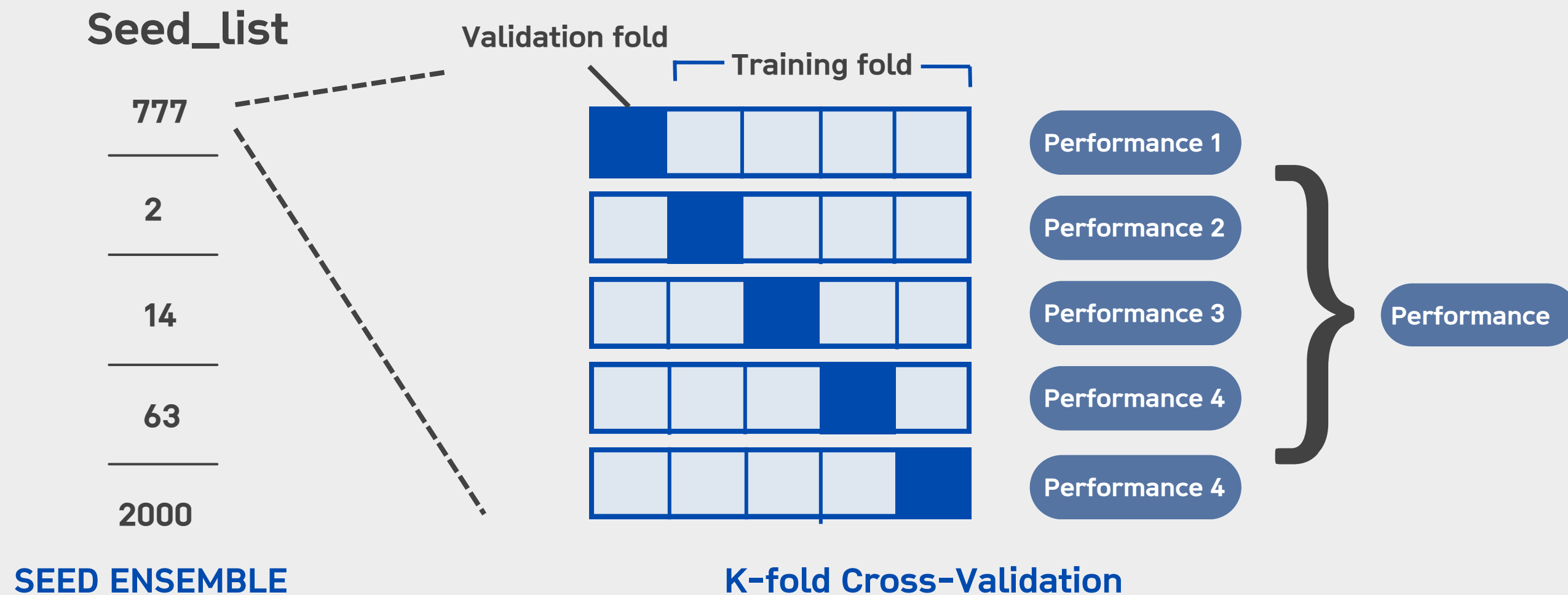
$\Rightarrow$ To generalize **model performance and prevent overfitting**

The mean and standard deviation of the validation scores for each fold were calculated, and the overall average across all folds was used to evaluate the validation performance of the experiment.

```
print('[Valid] ', end=' ')
y_pred = model.predict_proba(X_valid)[:,-1]
score = roc_auc_score(y_valid, y_pred)
print(f'Fold #{k_fold + 1} Score: {score:.4f}')
```

```
print(f'Avg. Score of validset: {np.mean(scores)}')
print(f'Std. Score of validset: {np.std(scores)}')
print()
```

* MODEL SELECTION AND SEED ENSEMBLE_ Model Architecture



- Only 50% of the test data's public score, sampled, can be checked, so focus was placed on overfitting and generalization of the model.
- **Seed Ensemble** was used to prevent overfitting by increasing model generalization and stability.
- Performance optimization experiments were conducted using **model ensembles and hyperparameter tuning** with three models: **XGB, LGBM, and CatBoost**.

* HYPERPARAMETER TUNING

OPTUNA

```
def objective(trial: optuna.Trial, config_params, df, target: str, base_features, cat_features) -> float:

    if config_params['model'] == 'XGB':
        params = {
            'objective': 'binary:logistic',
            'eval_metric': 'auc',
            'n_estimators': trial.suggest_int('n_estimators', 2200, 2400),
            'learning_rate': trial.suggest_float('learning_rate', 0.01, 0.03, log=True),
            'max_depth': trial.suggest_int('max_depth', 7, 8),
            'subsample': trial.suggest_float('subsample', 0.6, 0.9),
            'colsample_bytree': trial.suggest_float('colsample_bytree', 0.6, 0.9),
            'reg_lambda': trial.suggest_float('reg_lambda', 0.1, 1, log=True),
            'reg_alpha': trial.suggest_float('reg_alpha', 5, 9, log=True),
            'gamma': trial.suggest_float('gamma', 2.0, 3.0),
            'random_state': config_params['seed'],
        }
```

```
model_param_setting = {
    'XGB': {
        'n_estimators': 2357,
        'learning_rate': 0.016535403312726182,
        'max_depth': 7,
        'subsample': 0.6887677799128866,
        'colsample_bytree': 0.6879076928246035,
        'reg_lambda': 0.28211678046980465,
        'reg_alpha': 6.879354177483545,
        'gamma': 2.413362122265103,
        'objective': 'binary:logistic',
        'scale_pos_weight': 1,
        'verbosity': 0,
    },
}
```

Hyperparameter tuning of **XGBoost** was conducted using Optuna.

By considering parameters such as the number of trees, learning rate, maximum depth, sampling ratio, L1/L2 regularization, and gamma, the search space was gradually narrowed down to obtain the optimal hyperparameter set, which was then applied to the model.

* Final Model Selection

```
=====
Seed Number : 777
=====
<Model : XGB>
[Valid] Fold #1 Score: 0.7384
[Valid] Fold #2 Score: 0.7420
[Valid] Fold #3 Score: 0.7376
[Valid] Fold #4 Score: 0.7416
[Valid] Fold #5 Score: 0.7428
Avg. Score of validset: 0.7404880968025566
Std. Score of validset: 0.002063187912857618

=====
Seed Number : 2
=====
<Model : XGB>
[Valid] Fold #1 Score: 0.7420
[Valid] Fold #2 Score: 0.7388
[Valid] Fold #3 Score: 0.7392
[Valid] Fold #4 Score: 0.7423
[Valid] Fold #5 Score: 0.7380
Avg. Score of validset: 0.7400497815215938
Std. Score of validset: 0.00176950084224428
```

```
=====
Seed Number : 14
=====
<Model : XGB>
[Valid] Fold #1 Score: 0.7408
[Valid] Fold #2 Score: 0.7394
[Valid] Fold #3 Score: 0.7401
[Valid] Fold #4 Score: 0.7402
[Valid] Fold #5 Score: 0.7406
Avg. Score of validset: 0.7402477093606528
Std. Score of validset: 0.0004824819346555312

=====
Seed Number : 63
=====
<Model : XGB>
[Valid] Fold #1 Score: 0.7394
[Valid] Fold #2 Score: 0.7384
[Valid] Fold #3 Score: 0.7413
[Valid] Fold #4 Score: 0.7407
[Valid] Fold #5 Score: 0.7416
Avg. Score of validset: 0.7402954986096031
Std. Score of validset: 0.0012158551039369349
```

```
=====
Seed Number : 2000
=====
<Model : XGB>
[Valid] Fold #1 Score: 0.7381
[Valid] Fold #2 Score: 0.7406
[Valid] Fold #3 Score: 0.7423
[Valid] Fold #4 Score: 0.7395
[Valid] Fold #5 Score: 0.7406
Avg. Score of validset: 0.7402266975981797
Std. Score of validset: 0.0013729474099695947
```

After experimenting with various combinations including model ensembles, hyperparameter tuning, seed ensembles, and SMOTE-based oversampling, the final optimal model chosen was **a 5-seed ensemble of a single XGBoost model** with Optuna-based hyperparameter optimization applied.

#	팀	팀 멤버	최종점수	제출수
30	엄마손은 AI손		0.7422	129

THANK YOU

LG Almers 6th

엄마손은 AI손

고경준 김예진 나영은 이수민 정서윤

시술 시기 코드	Procedure Timing Code	시술 당시 나이	Age at Procedure	임신 시도 또는 마지막 임신 경과 연수	Pregnancy Attempt or Time Since Last Pregnancy
시술 유형	Treatment Type	특정 시술 유형	Specific Treatment Type	총 시술 횟수	Total Number of Treatments
클리닉 내 총 시술 횟수	Total Treatments at Clinic	IVF 시술 횟수	IVF Treatment Count	DI 시술 횟수	DI Treatment Count
총 임신 횟수	Total Number of Pregnancies	IVF 임신 횟수	IVF Pregnancy Count	DI 임신 횟수	DI Pregnancy Count
총 출산 횟수	Total Number of Births	IVF 출산 횟수	IVF Birth Count	DI 출산 횟수	DI Birth Count
단일 배아 이식 여부	Single Embryo Transfer Use	착상 전 유전 검사 사용 여부	Preimplantation Genetic Testing Use	착상 전 유전 진단 사용 여부	Preimplantation Genetic Diagnosis Use
배아 생성 주요 이유	Primary Reason for Embryo Creation	총 생성 배아 수	Total Embryos Created	미세주입된 난자 수	Number of Microinjected Oocytes
미세주입에서 생성된 배아 수	Number of Embryos Created via Microiniecton	이식된 배아 수	Number of Embryos Transferred	미세주입 배아 이식 수	Number of Microinjected Embryos Transferred
저장된 배아 수	Number of Stored Embryos	미세주입 후 저장된 배아 수	Number of Stored Embryos Post-Microiniecton	해동된 배아 수	Frozen-Thawed Embryos
해동 난자 수	Number of Frozen-Thawed Oocytes	수집된 신선 난자 수	Number of Freshly Retrieved Oocytes	저장된 신선 난자 수	Number of Stored Fresh Oocytes
혼합된 난자 수	Number of Mixed Oocytes	파트너 정자와 혼합된 난자 수	Number of Oocytes Mixed with Partner’s Sperm	기증자 정자와 혼합된 난자 수	Number of Oocytes Mixed with Donor Sperm
난자 출처	Source of Oocytes	정자 출처	Source of Sperm	난자 기증자 나이	Donor Oocyte Age
정자 기증자 나이	Donor Sperm Age	동결 배아 사용 여부	Frozen Embryo Use	신선 배아 사용 여부	Fresh Embryo Use
기증 배아 사용 여부	Donor Embryo Use	대리모 여부	Surrogacy Use	PGD 시술 여부	PGD Procedure Use
PGS 시술 여부	PGS Procedure Use	난자 채취 경과일	Days Elapsed Oocyte Retrieval	난자 해동 경과일	Days Elapsed Oocyte Thawing
난자 혼합 경과일	Days Elapsed Oocyte Mixing	배아 이식 경과일	Days Elapsed Embryo Transfer	배아 해동 경과일	Days Elapsed Embryo Thawing
남성 주 불임 원인	Primary Male Infertility Cause	남성 부 불임 원인	Secondary Male Infertility Cause	여성 주 불임 원인	Primary Female Infertility Cause
여성 부 불임 원인	Secondary Female Infertility Cause	부부 주 불임 원인	Primary Couple Infertility Cause	부부 부 불임 원인	Secondary Couple Infertility Cause
불명확 불임 원인	Unclear Infertility Cause	난관 질환	Tubal Disease	남성 요인	Male Factor
배란 장애	Ovulation Disorder	여성 요인	Female Factor	자궁경부 문제	Cervical Factor
자궁내막증	Endometriosis	정자 농도	Sperm Concentration	정자 면역학적 요인	Sperm Immunological Factor
정자 운동성	Sperm Motility	정자 형태	Sperm Morphology		